

The fundamental of chemical reactions: Acids and Bases

A few questions to ponder:

What is an acid?

What is a base?

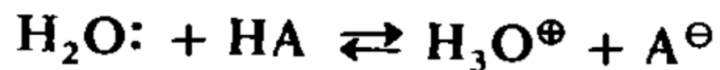
The acidity constant and pK_a

Note: Strength of acid and bases are always with reference to **water as solvent**

Strength of an acid (HA) is determined by the extent to which it can dissociate in water

In simple words, it is the ability or the ease with which it can donate a proton

The parameter used to describe is pK_a



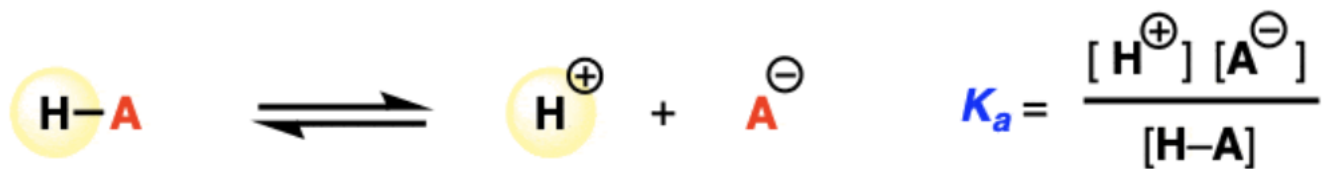
$$K_a \approx \frac{[\text{H}_3\text{O}^{\oplus}][\text{A}^{\ominus}]}{[\text{HA}]}$$

$$(pK_a = -\log_{10} K_a)$$

The smaller the value of pK_a , the stronger is the acid

The H_2O is in large excess. The H_2O concentration can be assumed to be negligible and taken as a constant

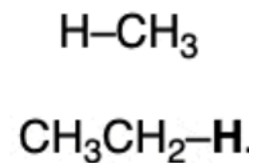
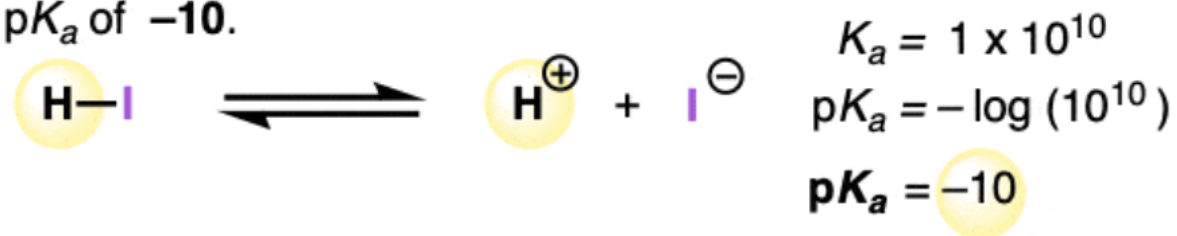
Strength of Acids



$$\text{p}K_a = -\log K_a$$

Why the minus sign?

In this way the experimentally determined K_a for H-I (K_a of 10^{10}) translates into a $\text{p}K_a$ of **-10**.



The acid dissociation constant K_a for H-OH (10^{-14}) becomes a $\text{p}K_a$ of **14** for H₂O

The acid dissociation constant K_a for the very weak acid H-CH₃ (10^{-50}) becomes a $\text{p}K_a$ of **50** for CH₄

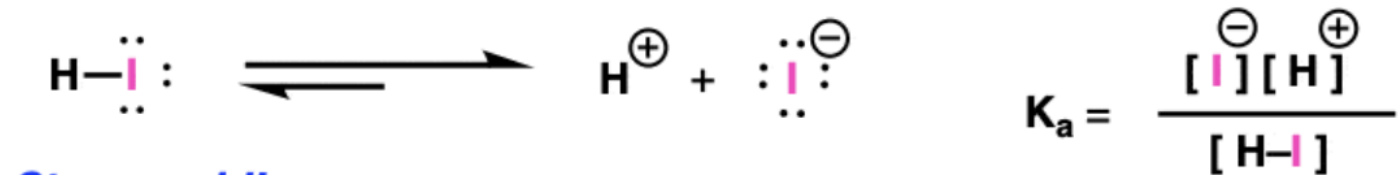
Factors that affect the acid strength (pK_a)

1. How strong is the HA bond?
2. Electronegativity of A
- 3. How stable is A^- as compared to HA**
4. Nature of the solvent
5. Temperature

Of these factor, the stability of A^- is often the most important in deciding strength of acid

Anions are stabilized by EWG (-I) and resonance (+R) effects as well

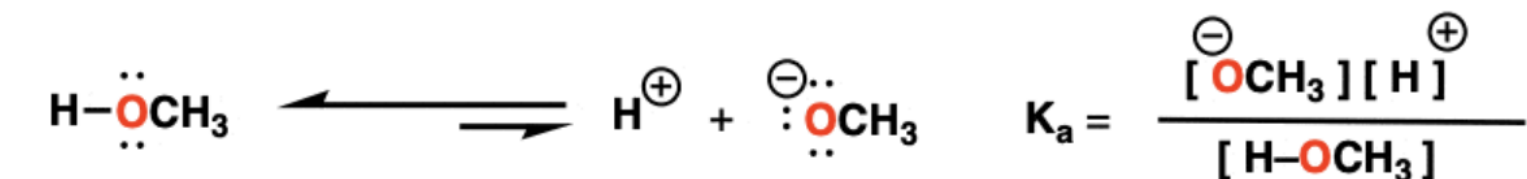
Compare is the anion is more stabilized than the acid



Strong acid!

experimental
measurement

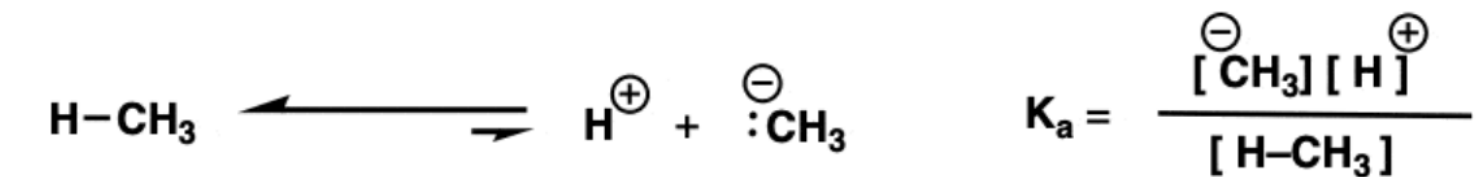
$$K_a = 1 \times 10^{10}$$



Weak acid

experimental
measurement

$$K_a = 3.2 \times 10^{-16}$$

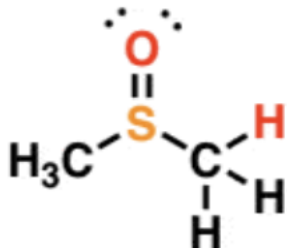
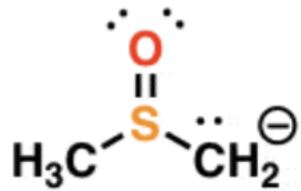
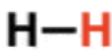
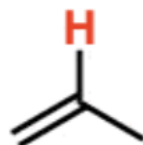
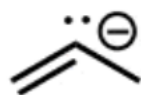
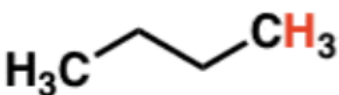
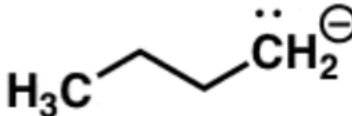


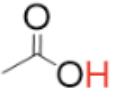
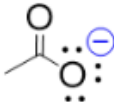
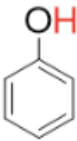
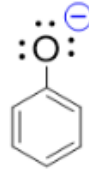
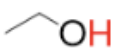
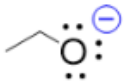
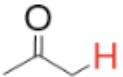
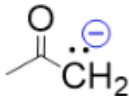
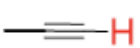
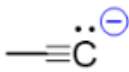
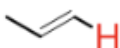
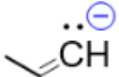
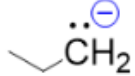
Extremely weak acid!

experimental
measurement

$$K_a \sim 1 \times 10^{-50}$$

Functional group	Example	pKa	Conjugate Base	Functional group	Example	pKa	Conjugate Base
Hydroiodic acid		-10		Thiols		13	
Hydrobromic acid		-9		Malonates		13	
Hydrochloric acid		-6		Water		14	
Sulfuric acid		-3		Alcohol		17	
Hydronium ion		-1.7		Ketone/aldehyde		20-24	
Sulfonic acids		-1		Nitrile		25	
Hydrofluoric acid		3.2		Ester		25	
Carboxylic acids		4		Alkyne		25	
Protonated amines		9-11		Amine		~35	

<i>Functional group</i>	<i>Example</i>	<i>pKa</i>	<i>Conjugate Base</i>
Sulfoxide		31	
Hydrogen		36	
Alkene		~43	
Alkane		~50	

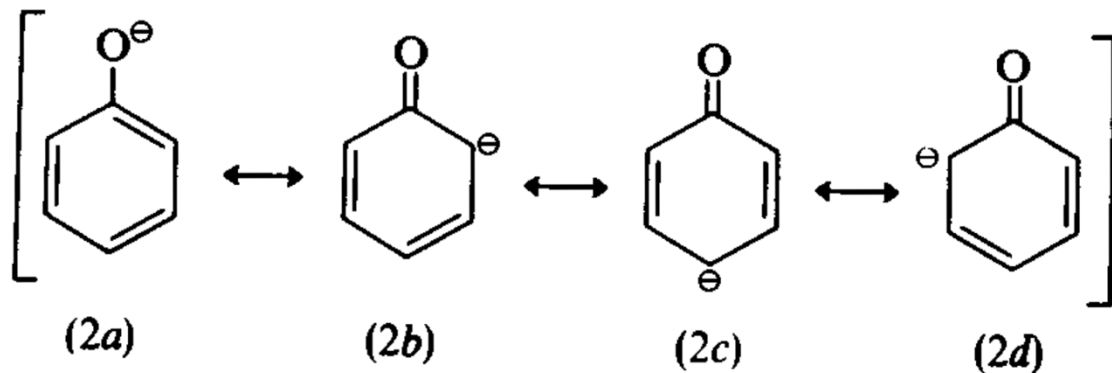
	Functional group	Example	pKa	Conjugate base	
Stronger acid	Hydrochloric acid	HCl	-7	$\text{:}\ddot{\text{Cl}}\text{:}^-$	Weaker base
	Carboxylic acid		5		
	Phenol		10		
	Alcohol		16		
	Ketone		19		
	Alkyne		25		
	Alkene		44		
Weaker acid	Alkane		51		Stronger base

Alkynes > Alkenes > Alkanes

Why?

Phenols are stronger acids than aliphatic alcohols

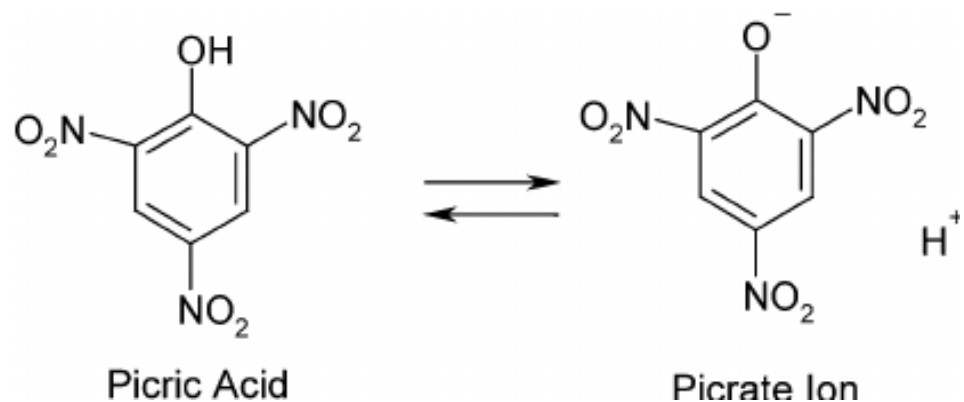
Alcohols and phenols are less acidic than carboxylic acids



Phenolate ions are stabilized by resonating structures

EWG makes acids stronger

Example – picric acid

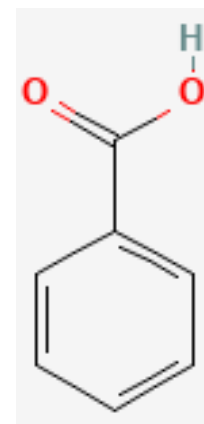
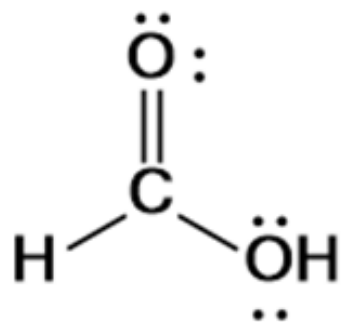


Though it is an alcohol, it is called an acid because its $pK_a \approx 0.3$ is almost comparable to carboxylic acids.

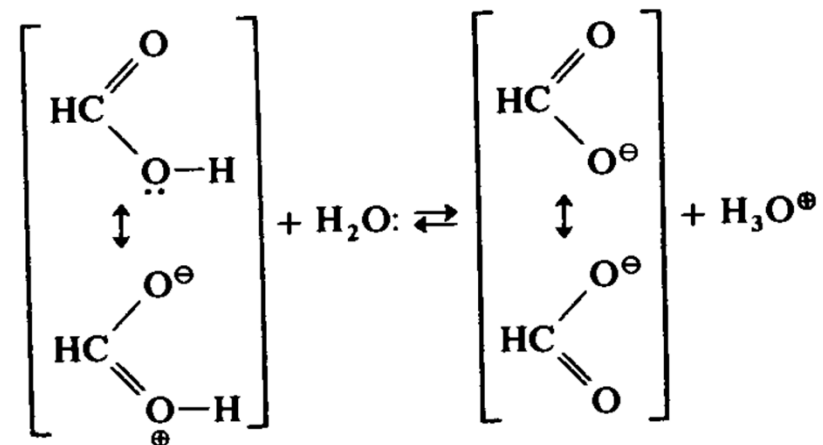
With the help of resonating structures, explain why picric acid is so acidic.

Predicting the trend of acidity

Formic acid vs Acetic acid vs Propanoic acid vs Benzoic acid



Resonating structures in formic acid and formate anion

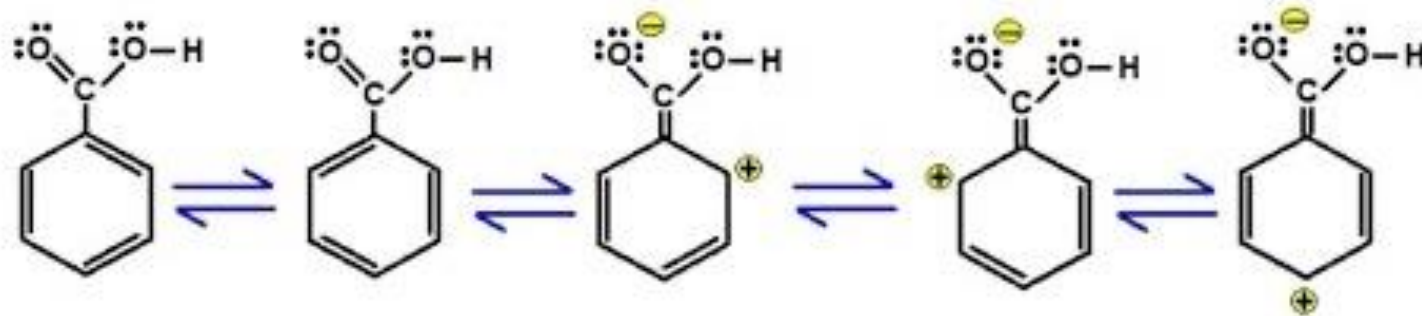
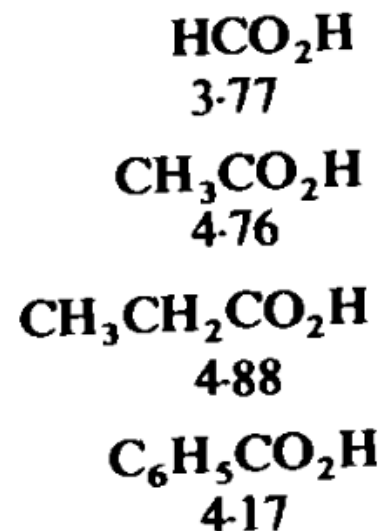


Note that both formic acid and formate anion has two resonating structures. But, in the formate anion there is actual charge delocalization, while in formic acid , there is only charge separation.

The resonating structures with actual charge delocalization are more stabilized, therefore the anion is more stabilized than the acid.

Strength of acids

Try to explain the pK_a ?

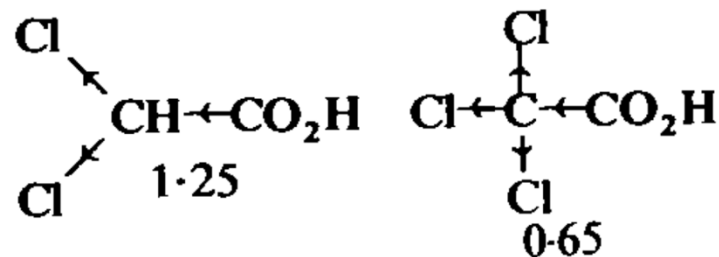
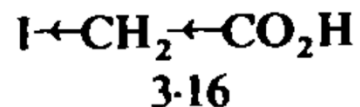
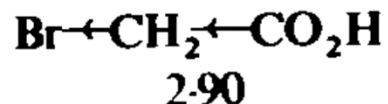
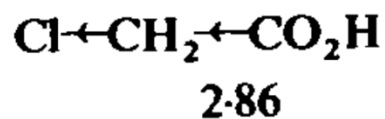
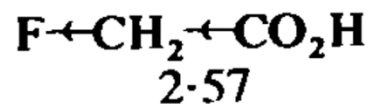
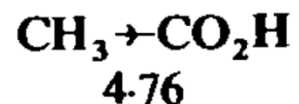


Note that formic acid (3.77) is more acidic than benzoic acid (4.17)

This is because the acid is more stabilized by resonance than the anion. The carbonyl oxygen cannot stabilize the anion because it is engaged in resonance with the benzene ring. Explain by drawing resonating structures.

Strength of acids

Try to explain the pK_a ?



Stability of anion

-I effect of adjacent electronegative group increases stability of anion

+I effect of methyl group destabilizes anion

Table 6.6. Equilibrium Acidities of Substituted Methanes in Dimethyl Sulfoxide^a

Compound	p <i>K</i>
CH ₃ NO ₂	17.2
CH ₃ COPh	24.7
CH ₃ COCH ₃	26.5
CH ₃ SO ₂ Ph	29.0
CH ₃ CO ₂ C ₂ H ₅	30.5 ^b
CH ₃ SO ₂ CH ₃	31.1
CH ₃ CN	31.3
CH ₃ CON(C ₂ H ₅) ₂	34.5 ^b

a. Except as noted otherwise, from W. S. Matthews, J. E. Bares, J. E. Bartmess, F. G. Bordwell, F. J. Cornforth, G. E. Drucker, Z. Margolin, R. J. McCallum, G.J McCollum, and N. R. Vanier, *J. Am. Chem. Soc.*, **97**, 7006 (1975).

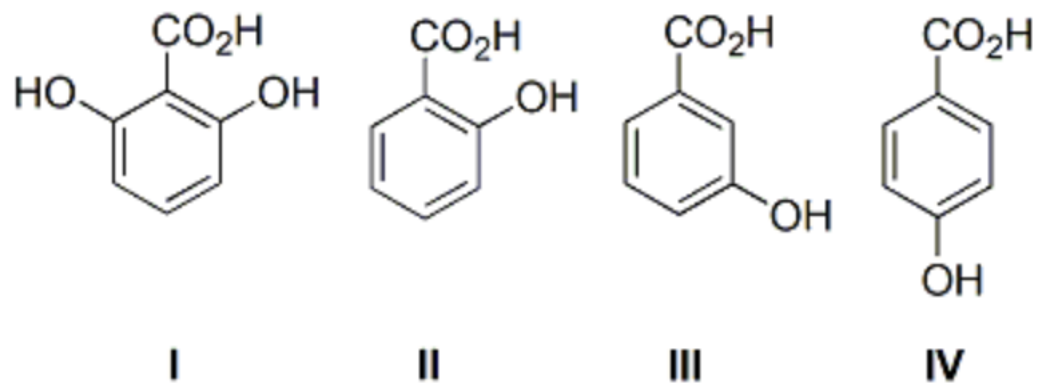
b. F. G. Bordwell and H. E. Fried, *J. Org. Chem.*, **46**, 4327 (1981).

EWG order

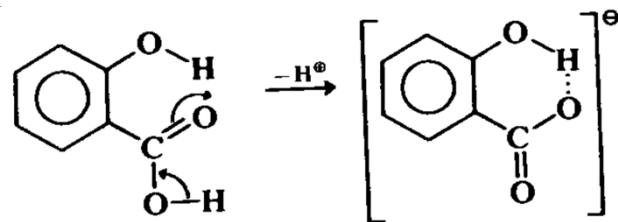


Strength of aromatic acids

Stabilization of anion by **Hydrogen Bonding** and **-I effects**



Order of acidity: **I > II > III > IV**



Salicylic acid (II) is more acidic than p-hydroxy benzoic acid (IV) because of H-Bonding

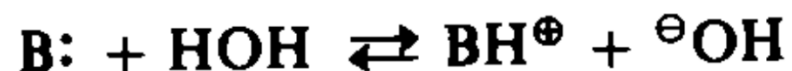
In I, two OH group participates in H-Bonding (also -I effect of OH)

In III vs IV, negative inductive (-I) effect of OH group is greater in meta than in para, hence III anion is more stabilized

Factors affecting strength of base (pK_b)

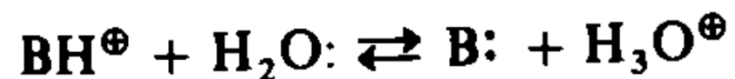
Strength: How easily can it grab a proton?

How stable is the cation (BH^+)?



$$K_b \approx \frac{[BH^{\oplus}][{}^{\ominus}OH]}{[B:]}$$

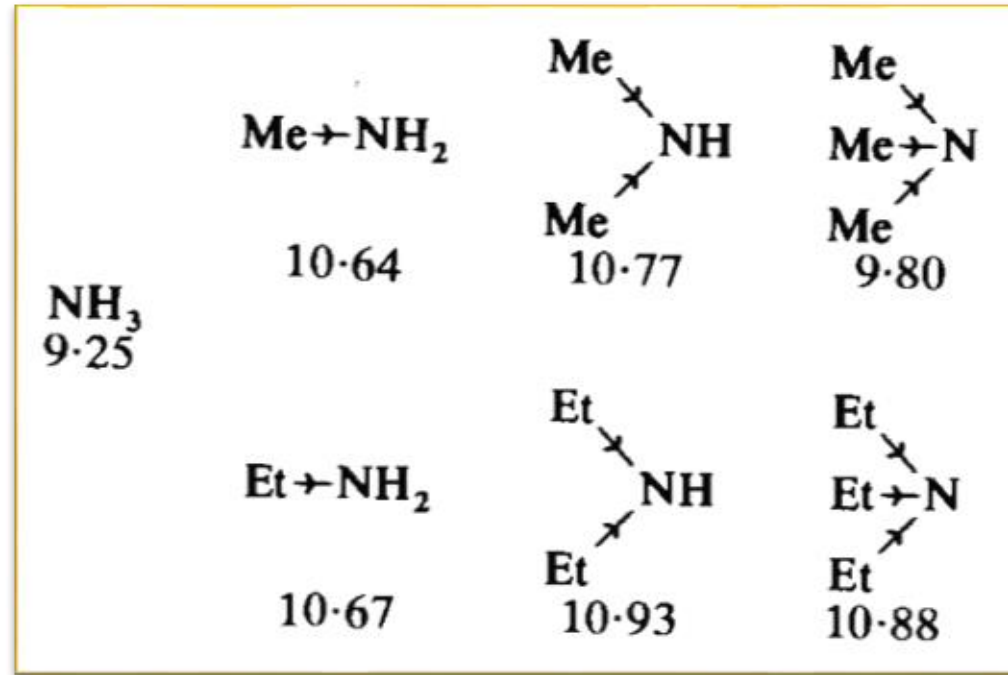
Note: The term pK_a is also used to denote the strength of bases (conjugated acid)



$$K_a \approx \frac{[B:][H_3O^{\oplus}]}{[BH^{\oplus}]}$$

Important note: Smaller the pK_a , weaker is the base (unlike acids)

Basicity in substituted amines



Ammonia < 1° < 2° ≈ or > 3°

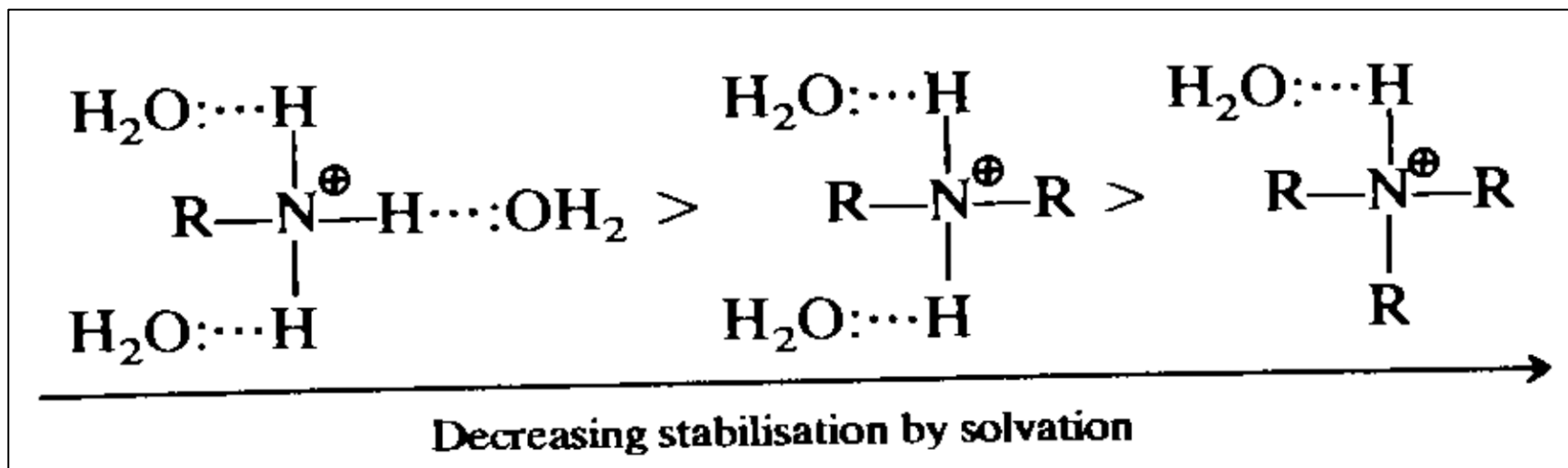
Factors to be considered for stability of cation BH^+

1. Inductive effect

2. Solvation effect (Steric effects)

In solvent phase, solvent effects overrides inductive effect for 3° amines
In gaseous phase, inductive effect dominates

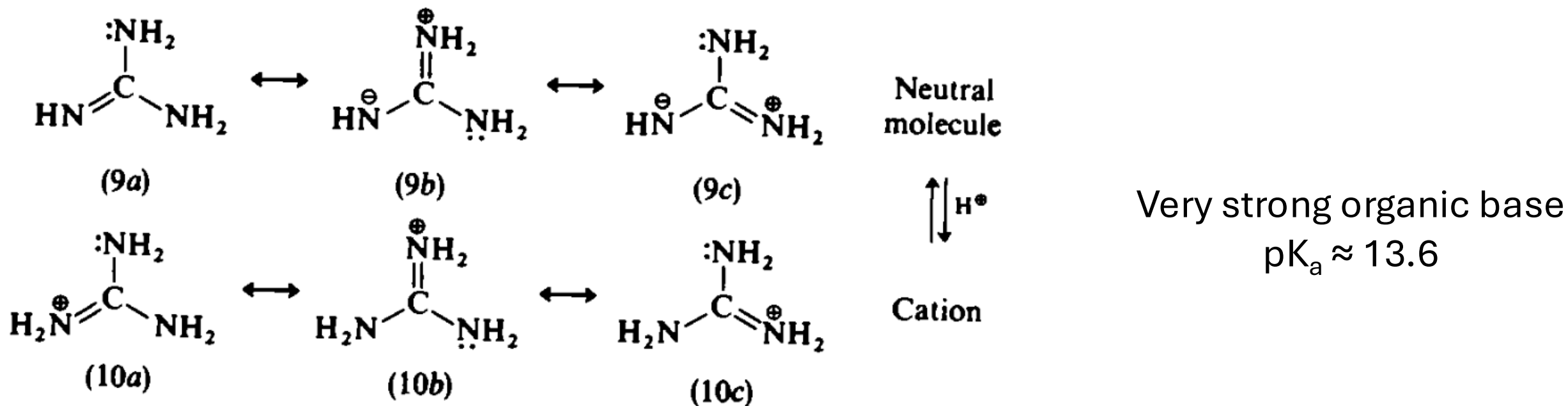
Predicting the basicity of amines



Trimethyl amine is less basic than methyl amine and almost of equal strength with dimethyl amine. Explain why?

+I effect of methyl groups increases stability of the cation. Therefore, order of strength is expected to be $\text{NH}_3 < 1^\circ < 2^\circ < 3^\circ$. But going from secondary to tertiary, the reverse order is observed (same case for ethyl group). Why? Because of solvation effects (hydrogen bonding). The more the bulky is the cation, less is the stabilization by hydration and therefore order of basicity is $\text{NH}_3 < 1^\circ < 2^\circ \approx \text{or} > 3^\circ$

Guanidine: One of the strongest organic bases



Both the neutral molecule (9a) and the cation formed after accepting proton (10a) is stabilized by resonance. Both 9 and 10 has three resonating structures (a,b,c). But, in case of 10 (cation), there is **actual charge delocalization**. The positive charge is **delocalized symmetrically and spread over three equivalent nitrogen atoms**. Therefore, the cation is more stabilized than the unprotonated base, **making protonation “energetically profitable”** and guanidine an extremely strong base

Buffers

- Can resist pH change upon the addition of an acidic or basic components.
- Able to neutralize small amounts of added acid or base, thus maintaining a stable pH of the solution
- A working pH range -- how much acid/base can be neutralized before pH changes
- A capacity -- amount by which pH will change

Preparation of buffers

- mixing a large volume of a weak acid with its conjugate base (eg. acetic acid – acetate ion, $\text{CH}_3\text{COOH} - \text{CH}_3\text{COO}^-$)
- mixing a large volume of weak base with its conjugate acid (eg. ammonia – ammonium ion, $\text{NH}_3 - \text{NH}_4^+$)

Phosphate buffer is a very commonly used buffer in research labs. It falls under the second category. It's made up of a weak base (HPO_4^{2-}) and its conjugate acid (H_2PO_4^-). The pH of a phosphate buffer is usually maintained at a physiological pH of 7.4.

How does a buffer work?

Le Chatelier's Principle

Le Chatelier's principle states that if a **dynamic equilibrium** is disturbed by changing the conditions, the position of equilibrium shifts to counteract the change to reestablish an equilibrium

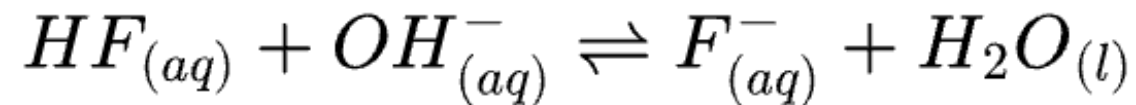
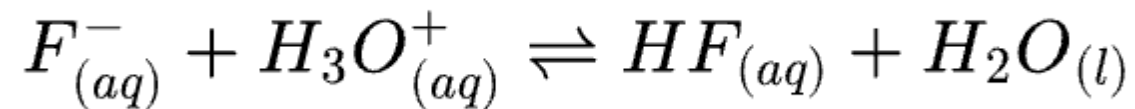
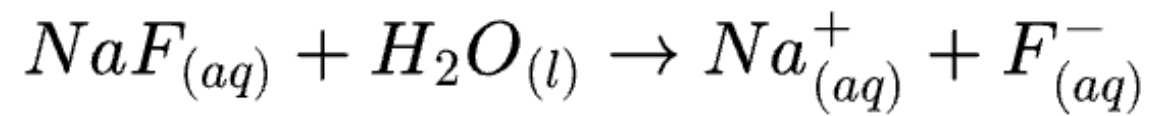
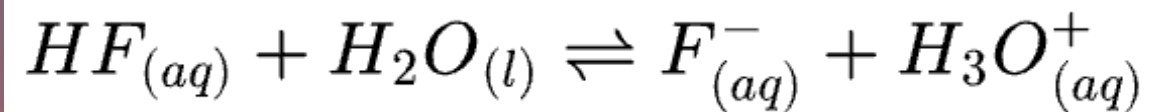


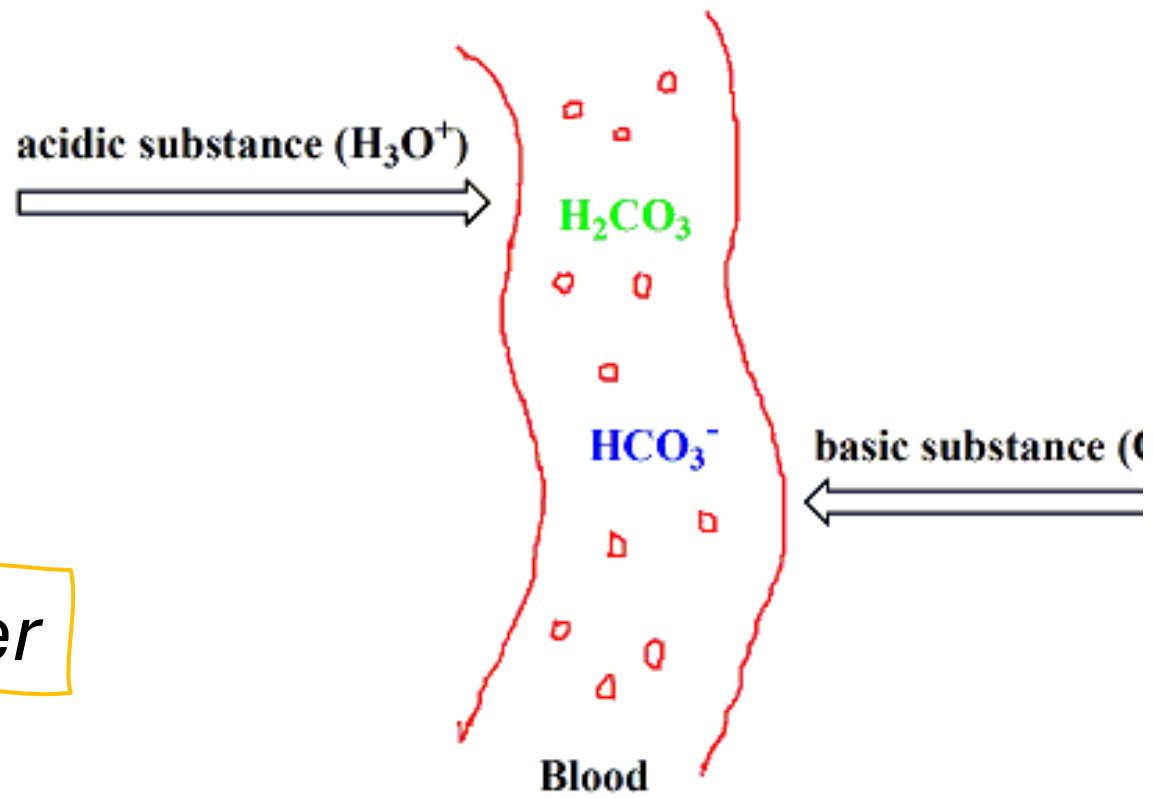
The position of equilibrium moves to the right if you increase the concentration of A.



The position of equilibrium moves to the left if you decrease the concentration of A.

Choosing buffer components





Blood is a buffer